
Entra ID to JFrog SCIM Provisioning

Behaviour Validation, Test Results & Migration Recommendations

JFrog Professional Services | SSO + SCIM Enablement

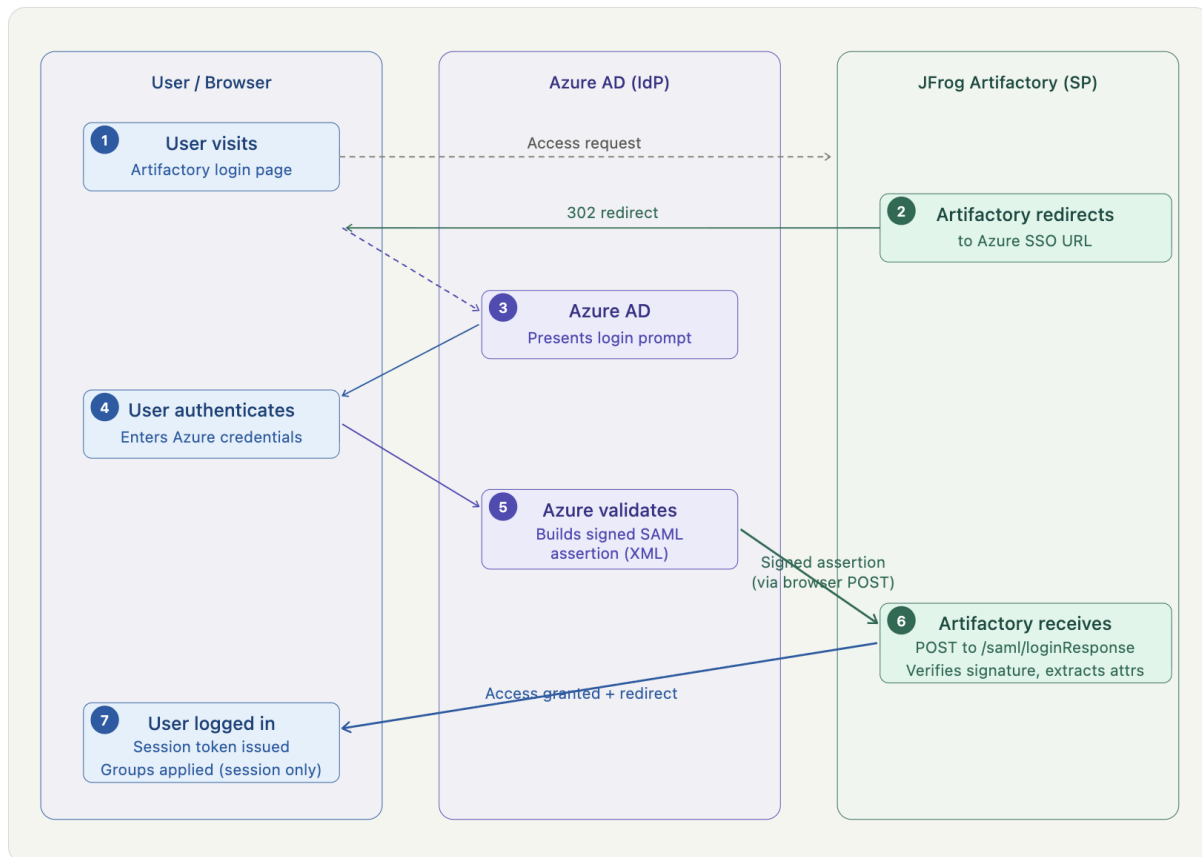
1 SAML and SCIM

What is SAML?

Security Assertion Markup Language (SAML 2.0) is an open standard that lets an Identity Provider (IdP) — such as Azure Active Directory — tell a Service Provider (SP) — such as JFrog Artifactory — that a user has been authenticated.

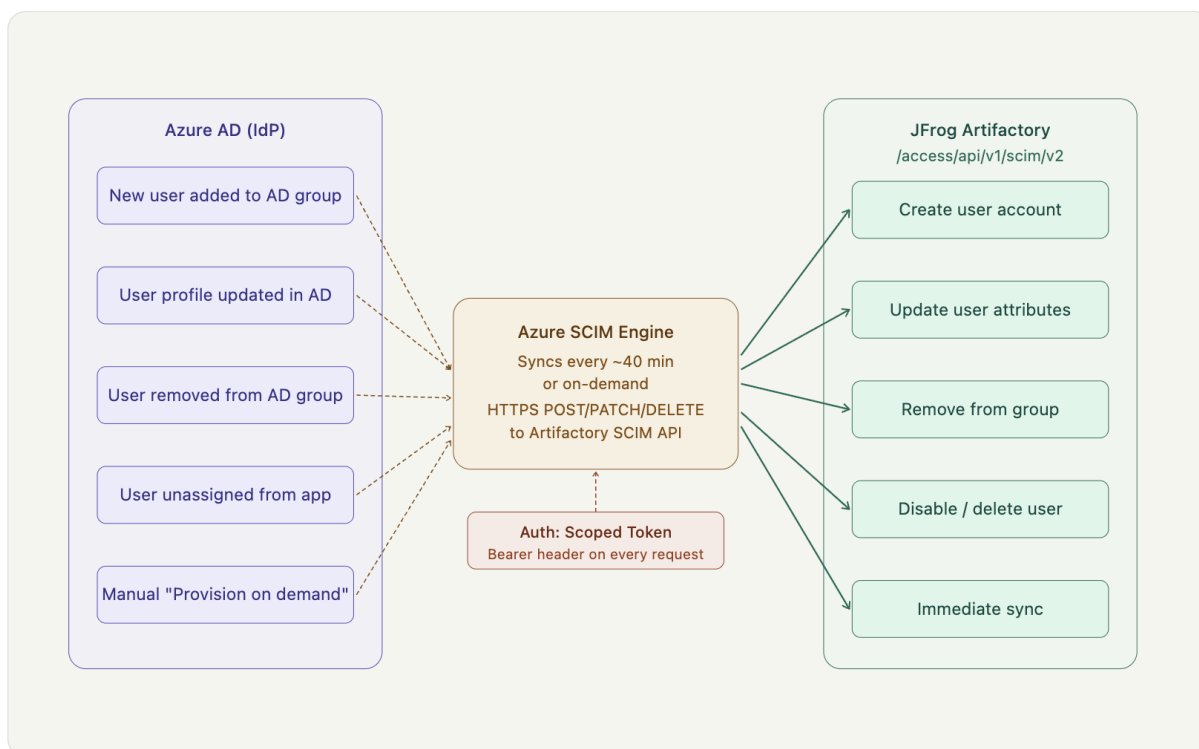
The user never sends their password to Artifactory. They prove identity once to Azure AD, and Azure AD gives Artifactory a signed XML voucher called a SAML Assertion.

SAML authentication flow



What is SCIM?

System for Cross-domain Identity Management (SCIM 2.0) is a REST-based protocol for automating user and group lifecycle management. While SAML handles authentication (login), SCIM handles provisioning — creating, updating, and disabling accounts in Artifactory whenever changes happen in Azure AD.



Overview

This document validates how SCIM provisioning from Microsoft Entra ID behaves against a JFrog Platform instance that already contains locally-created (internal) groups and users. It records the recommended migration approaches first, followed by the executed test matrix with supporting evidence.

Two matching rules govern all observed behaviour and are worth stating up front: JFrog matches groups by display name (a same-named group is adopted, not duplicated), and JFrog matches users by username, not by email address. SCIM group-membership operations are incremental, so pre-existing local members are preserved on a normal sync.

2 Migration Recommendations

Two viable approaches exist for bringing existing local groups under Entra/SCIM management. The choice can be made per group.

Option A — Same group name (adopt in place)

Recreate the existing local groups in Entra under the same names, so SCIM adopts each group in place and preserves its name and permission targets. Follow these steps:

Step 1: Collect. Identify every JFrog local group and the users attached to each.

Step 2: Recreate in Entra. Create the same group names in Entra ID and attach the same users, ensuring the SCIM username matches the existing JFrog username for each user.

Step 3: Configure SCIM. Assign the groups to the enterprise application and enable SCIM provisioning.

Step 4: Verify sync. Trigger a sync (provision on demand or the scheduled cycle) and confirm the groups and users appear correctly in JFrog.

Step 5: Test. Validate the scenarios (membership, attribute updates, de-provisioning) against the expected behaviour.

- **Pros:** Least permission rework — existing group names and their permission targets are retained.
- **Cautions:** Requires exact username alignment; privileged groups inherit their existing roles (including Platform Administrator) and must be reviewed before adoption.

Because users are matched by username, the SCIM username attribute sent by Entra must resolve to the existing platform usernames. This is an attribute-mapping decision, not a manual re-creation of accounts. If the values differ, a separate provisioned account is created alongside the existing one.

Option B — New group name (provision fresh, repoint permissions)

Create new group names in Entra so SCIM provisions them as brand-new externally-managed groups, then repoint permission targets from the old local group to the new group. Follow these steps:

Step 1: Collect. Identify every JFrog local group, its users, and the permission targets that reference it.

Step 2: Create in Entra. Create new group names in Entra ID and attach the intended users, ensuring the SCIM username matches the existing JFrog username for each user.

Step 3: Configure SCIM. Assign the new groups to the enterprise application and enable SCIM provisioning.

Step 4: Verify sync. Trigger a sync and confirm the new groups and users are provisioned correctly in JFrog.

Step 5: Repoint permissions. Add the new group to every permission target that referenced the old group, verify access, then remove the old group.

Step 6: Test. Validate the scenarios (membership, attribute updates, de-provisioning) against the expected behaviour.

- **Pros:** Avoids username-collision and role-inheritance ambiguity; **a clean, greenfield SCIM footprint.**
- **Cautions:** Every permission target and repository assignment referencing the old group must be repointed to the new group, with a carefully sequenced cutover.

Recommended cutover sequence for Option B: provision the new group → verify membership → add the new group to all relevant permission targets alongside the old group → verify access → only then remove the old group. Never remove the old group's permissions before the new group is provisioned and verified.

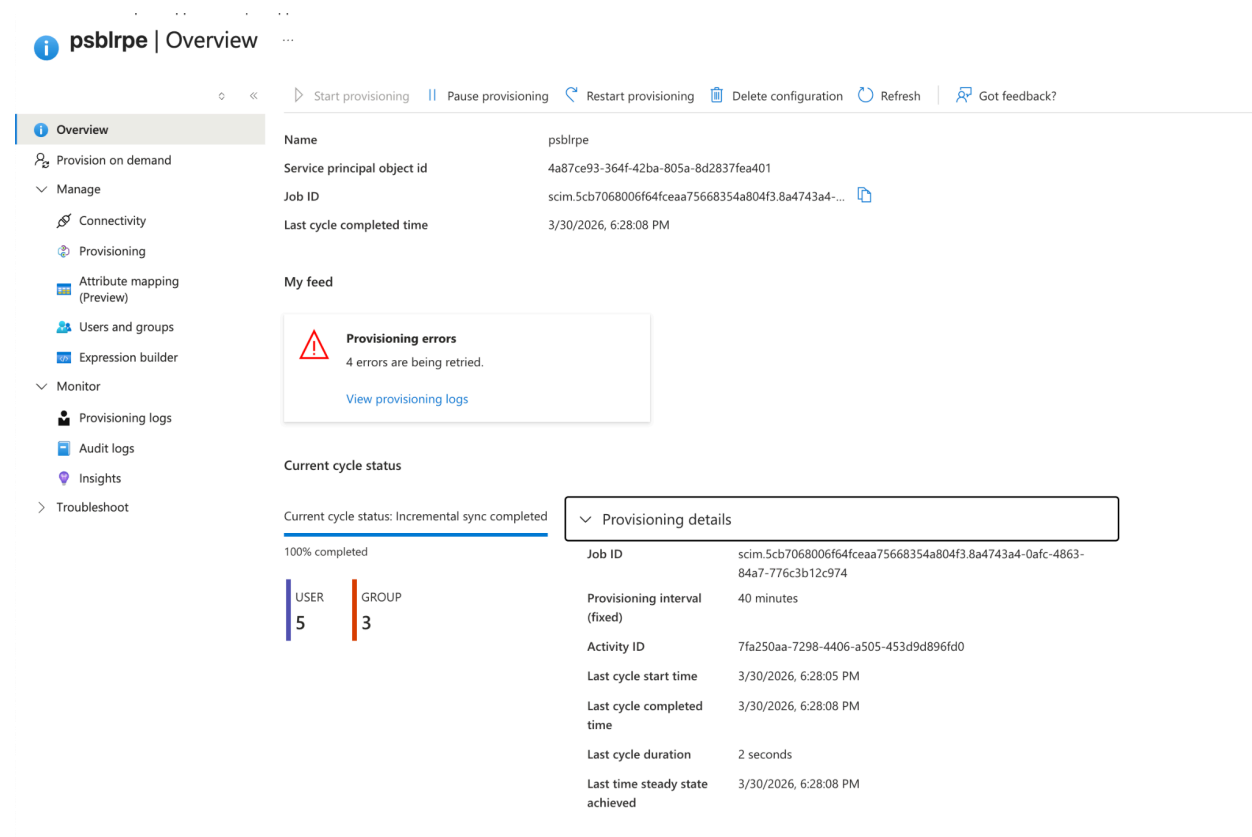
Alignment prerequisite (both options): the SCIM username attribute and the SAML NameID must resolve to the same value (typically both userPrincipalName, or both email), or a user will hold a provisioned account carrying group access separate from the login account that carries none.

3 SCIM Setup (Azure AD)

This section details the one-time configuration referenced by the “Configure SCIM” step in the migration options above.

SCIM setup steps (Azure AD)

- Step 1:** Create a new Enterprise Application in Azure AD (non-gallery — the existing JFrog gallery template cannot be used for SCIM).
- Step 2:** Go to the application → Provisioning → Get Started.
- Step 3:** In Artifactory, navigate to Administration → Authentication → SCIM → Generate Token.
- Step 4:** Copy the Base URL and token from Artifactory.
- Step 5:** In Azure, set the Tenant URL to `https://<Artifactory_URL>/access/api/v1/scim/v2` and the Secret Token to the scoped token from Artifactory.
- Step 6:** Click Test Connection — a green success notification confirms connectivity.
- Step 7:** Set provisioning mode to Automatic and save.



The screenshot shows the Azure AD Provisioning interface for an application named 'psblrpe'. The left sidebar contains navigation links: Overview, Provision on demand, Manage, Connectivity, Provisioning, Attribute mapping (Preview), Users and groups, Expression builder, Monitor, Provisioning logs, Audit logs, Insights, and Troubleshoot. The main content area displays the following information:

- Overview:** Name: psblrpe; Service principal object id: 4a87ce93-364f-42ba-805a-8d2837fea401; Job ID: scim.5cb7068006f64fcea75668354a804f3.8a4743a4-...; Last cycle completed time: 3/30/2026, 6:28:08 PM.
- My feed:** A warning icon indicates 'Provisioning errors' with the message '4 errors are being retried.' and a link to 'View provisioning logs'.
- Current cycle status:** Current cycle status: Incremental sync completed. A progress bar shows 100% completed. A summary shows 5 users and 3 groups.
- Provisioning details:**
 - Job ID: scim.5cb7068006f64fcea75668354a804f3.8a4743a4-0afc-4863-84a7-776c3b12c974
 - Provisioning interval (fixed): 40 minutes
 - Activity ID: 7fa250aa-7298-4406-a505-453d9d896fd0
 - Last cycle start time: 3/30/2026, 6:28:05 PM
 - Last cycle completed time: 3/30/2026, 6:28:08 PM
 - Last cycle duration: 2 seconds
 - Last time steady state achieved: 3/30/2026, 6:28:08 PM

Connectivity — the JFrog Platform SCIM Tenant URL and token are entered and tested.

psblrpe | Connectivity

Overview

Provision on demand

Manage

Connectivity

Provisioning

Attribute mapping (Preview)

Users and groups

Expression builder

Monitor

Troubleshoot

Save

Discard

Got feedback?

This is a new version of the provisioning user experience. You can provide us feedback and suggestions on the new user experience using the "Got feedback" button. [Click here to switch to the legacy experience.](#)

Microsoft Entra needs the following information to connect to the API for 'psblrpe' and synchronize user data. A successful connection test is required before saving.

Select authentication method:
Bearer authentication

Tenant URL ⓘ *
<https://psblrjfrog.io/access/api/v1/scim/v2>

Secret token
•

Test connection

Provisioning — provisioning mode set to Automatic, provisioning status On.

psblrpe | Provisioning

Overview

Provision on demand

Manage

Connectivity

Provisioning

Attribute mapping (Preview)

Users and groups

Expression builder

Monitor

Troubleshoot

Save

Discard

Provisioning Mode
Automatic

Use Microsoft Entra to manage the creation and synchronization of user accounts in psblrpe based on user and group assignment.

Admin Credentials

Mappings

Settings

Provisioning Status ⓘ
On Off

Home > Enterprise applications | All applications > psblrpe

psblrpe | Attribute mapping (Preview) ...

◊ << >> Restore default mappings

Overview

Provision on demand

Manage

Connectivity

Provisioning

Attribute mapping (Preview)

Users and groups

Expression builder

> Monitor

> Troubleshoot

Mappings allow you to define how data should flow between Microsoft Entra ID and customappsso.

Name	Enabled
Provision Microsoft Entra ID Groups	Yes
Provision Microsoft Entra ID Users	Yes

Attribute mapping — Entra ID Users and Groups provisioning mappings enabled.

Synchronization trigger

- **Default behaviour:** Azure AD provisioning runs on a background service and typically triggers every 40 minutes; larger tenants may see variation.
- **Manual trigger:** Provision on demand syncs a single user or group immediately for testing.

Home > Enterprise applications | All applications > psblrpe

psblrpe | Users and groups ...

◊ << + Add user/group Edit assignment Remove assignment Update credential Refresh Manage view Got feedback?

Overview

Provision on demand

Manage

Connectivity

Provisioning

Attribute mapping (Preview)

Users and groups

Expression builder

> Monitor

> Troubleshoot

The application will appear for assigned users within My Apps. Set 'visible to users' to no in properties to prevent this.

Assign users and groups to app-roles for your application here. To create new app-roles for this application, use the [application registration](#)

First 200 shown, search all users & groups

	Display name	Object type	Role assigned
☐	 Anil Kumar Thatavarthi	User	User
☐	 Giridhar Naga	User	User
☐	 Jayanth Suresh	User	User
☐	 Sumanth S V	User	User
☐	 ps-group-test	Group	User
☐	 temasek-admin-group	Group	User
☐	 temasek-scm-group	Group	User

Home > Enterprise applications | All applications > psblrpe

psblrpe | Provisioning logs

Download
Refresh
Manage view
Got feedback?

Search Identity
Add filter

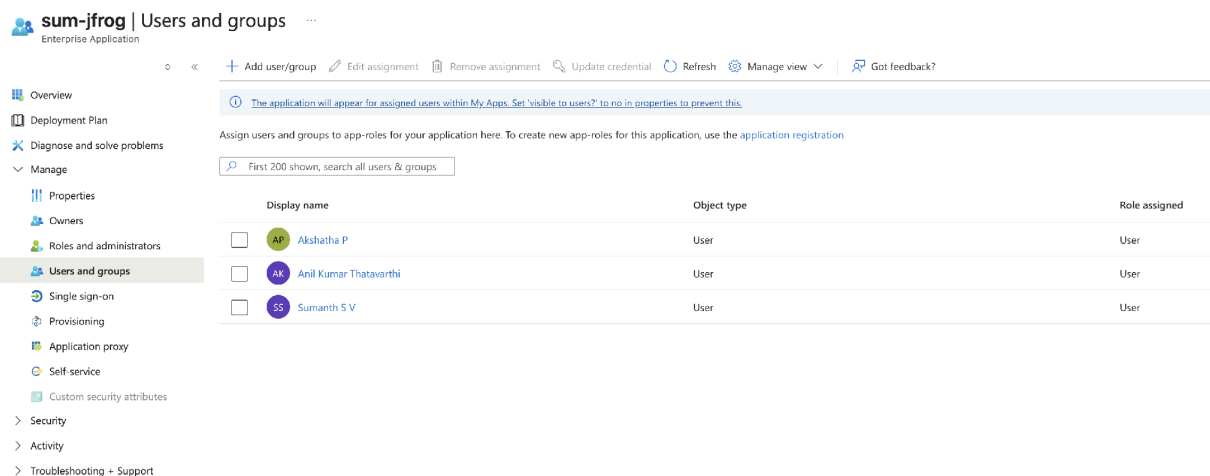
Show dates as: Local
Date range: Last 24 hours
Action: All
Status: All
Application contains 4a87ce93-364f-42ba-805a-8d2837ea401
Reset filters

Date	Identity	Action	Source system	Target system	Status	Application
3/30/26, 1:01:04 PM	Display name Sahar Noiman Source ID f1c5680d-1158-435a-a93a-3303afb22c7e	Create	Microsoft Entra ID	customappsso	Skipped	Display name p Object ID 4a87f
3/30/26, 11:04:41 AM	Display name temasek-scm-group Source ID 7490bf4-50b2-49ad-8089-a811fad8dab Target ID temasek-scm-group	Update	Microsoft Entra ID	customappsso	Success	Display name p Object ID 4a87f
3/30/26, 11:04:39 AM	Display name temasek-scm-group Source ID 7490bf4-50b2-49ad-8089-a811fad8dab Target ID temasek-scm-group	Update	Microsoft Entra ID	customappsso	Success	Display name p Object ID 4a87f
3/30/26, 11:04:15 AM	Display name temasek-scm-group Source ID 7490bf4-50b2-49ad-8089-a811fad8dab Target ID temasek-scm-group	Update	Microsoft Entra ID	customappsso	Success	Display name p Object ID 4a87f
3/30/26, 11:04:15 AM	Display name temasek-scm-group Source ID 7490bf4-50b2-49ad-8089-a811fad8dab Target ID temasek-scm-group	Update	Microsoft Entra ID	customappsso	Success	Display name p Object ID 4a87f
3/30/26, 11:04:13 AM	Display name temasek-scm-group Source ID 7490bf4-50b2-49ad-8089-a811fad8dab Target ID temasek-scm-group	Update	Microsoft Entra ID	customappsso	Success	Display name p Object ID 4a87f
3/30/26, 11:04:10 AM	Display name Sumanth S V Source ID 62a3a425-4492-4fa3-8291-61066f10840 Target ID sumanthsv@jfrog.com	Update	Microsoft Entra ID	customappsso	Failure	Display name p Object ID 4a87f
3/30/26, 11:04:09 AM	Display name Giridhar Naga Source ID 4cd523d1-db5e-49fb-9a69-5105e5c9b72f Target ID nagag@jfrog.com	Update	Microsoft Entra ID	customappsso	Success	Display name p Object ID 4a87f
3/30/26, 11:04:09 AM	Display name Sumanth S V	Update	Microsoft Entra ID	customappsso	Success	Display name p

Users and groups synced, and the provisioning logs showing per-object sync results.

4 Test Environment

Testing was performed against a live JFrog Platform (SaaS) instance integrated with Microsoft Entra ID via a SAML enterprise application for single sign-on and the JFrog SCIM connector for provisioning. Groups were assigned to the enterprise application in Entra; only in-scope (assigned) identities are provisioned. Assigning a group to an enterprise application requires an Entra ID P1 or P2 license.



sum-jfrog | Users and groups ...

Enterprise Application

◊ « + Add user/group Edit assignment Remove assignment Update credential Refresh Manage view Got feedback?

Overview

Deployment Plan

Diagnose and solve problems

Manage

Properties

Owners

Roles and administrators

Users and groups

Single sign-on

Provisioning

Application proxy

Self-service

Custom security attributes

> Security

> Activity

> Troubleshooting + Support

① The application will appear for assigned users within My Apps. Set 'visible to users?' to no in properties to prevent this.

Assign users and groups to app-roles for your application here. To create new app-roles for this application, use the [application registration](#)

First 200 shown, search all users & groups

	Display name	Object type	Role assigned
☐	AP Akshatha P	User	User
☐	AK Anil Kumar Thatavarthi	User	User
☐	SS Sumanth S V	User	User

Entra enterprise application — Users and groups assignment blade.

4 Test Results

4.1 Provisioning Connectivity

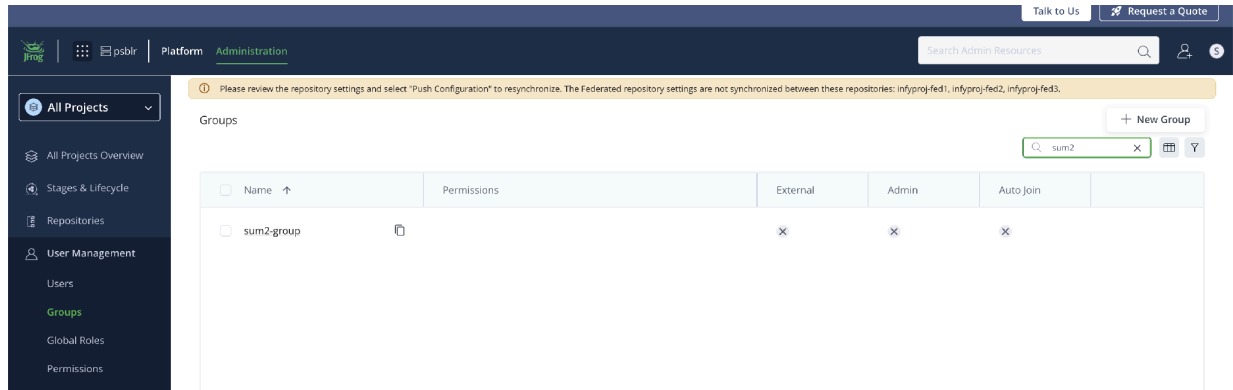
ID	Scenario	Expected Result	Pass	Result / Notes
1.1	Initial SCIM handshake	Provisioning starts without errors using the Base URL and secret token.	Pass	Handshake succeeded; provisioning enabled without errors.
1.2	Scheduled interval	Verify the next-cycle time in Entra. Entra syncs on a fixed service interval.	Pass	Confirmed. Default interval is 40 minutes.
1.3	Manual trigger	"Provision on demand" immediately syncs a selected user or group.	Pass	Confirmed. On-demand group and user provisioning reflected in the JFrog Platform.

4.2 User & Group Management

ID	Scenario	Expected Result	Pass	Result / Notes
2.1	Existing data integrity	Existing local users remain active and are not deleted when SCIM is enabled.	Pass	Confirmed. SCIM updates only the users and groups assigned to the enterprise application; no other entities are affected.
2.2	Conflict resolution (same-name group)	Where a group of the same name exists on both sides, Entra manages the membership list in JFrog.	Pass	Confirmed. The same-named group is adopted; existing local members are retained while new members from the Entra group are added (union of both sides).
2.3	Attribute sync	Changing a user's display name or email in Entra updates the corresponding fields in JFrog.	Pass	Confirmed. A manually changed email in the platform was overwritten by the Entra value on the next sync.
2.4	New user provisioning	Adding a new user to the Entra group creates that account in JFrog automatically.	Pass	Confirmed. The user was created and added to the group on the next sync.

Evidence — same-name group adoption and membership union:

Local group before provisioning — External and Admin unset.



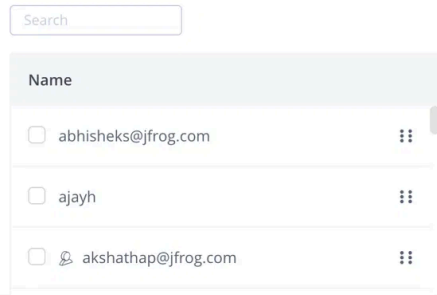
The screenshot shows the JFrog Administration console. On the left is a sidebar with navigation links: All Projects, All Projects Overview, Stages & Lifecycle, Repositories, User Management, Users, Groups (highlighted), Global Roles, and Permissions. The main content area is titled 'Groups' and contains a table with the following data:

Name	Permissions	External	Admin	Auto Join
sum2-group		x	x	x

At the top of the 'Groups' section, there is a search bar with 'sum2' entered and a '+ New Group' button. A yellow warning banner at the top of the main content area reads: 'Please review the repository settings and select "Push Configuration" to resynchronize. The Federated repository settings are not synchronized between these repositories: infyproj-fed1, infyproj-fed2, infyproj-fed3.'

Users

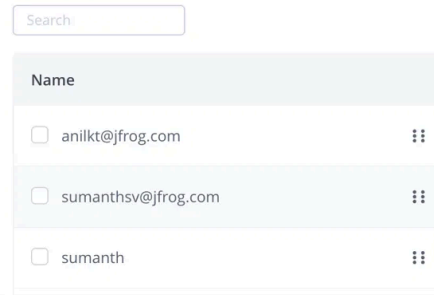
174 Available Users



The '174 Available Users' section includes a search bar and a table with the following data:

Name
☐ abhisheks@jfrog.com
☐ ajayh
☐ akshathap@jfrog.com

3 Selected Users



The '3 Selected Users' section includes a search bar and a table with the following data:

Name
☐ anilkt@jfrog.com
☐ sumanthsv@jfrog.com
☐ sumanth

Same group after provisioning — now flagged External (SCIM-managed).



Name	Permissions	External	Admin	Auto Join
sum2-group		✓	x	x

 This group is automatically updated using SCIM. Any change to the group might be lost in the next automatic update to the group.

Group Settings

* Group Name	Description	External ID 
sum2-group		fd71f273-faa1-44cf-abab-3d4dc4b6e3b6

Users

173 Available Users

Name
☐ abhisheks@jfrog.com
☐ ajayh
☐ alice-dev
☐ alpha-slvr-a
☐ alpha-slvr-b



5 Selected Users

Name
☐ anikt@jfrog.com
☐ sumanthsv@jfrog.com
☐ akshathap@jfrog.com
☐ jeevanr@jfrog.com
☐ sumanth

Group Permissions

Membership after provisioning — the union of local and Entra members is retained.

Name ↑	Email	Realm	Groups	Admin	St
 akshathap@jfr...	akshathap@jfrog.com	scim	1 sum-group		

A provisioned user appears in the scim realm, added to the adopted group.

Note — role inheritance. A same-named local group that already carries a role (for example Platform Administrator) passes that role to every user provisioned into it. Review privileged groups before adopting them.

4.3 Membership & De-provisioning

ID	Scenario	Expected Result	Pass	Result / Notes
3.1	User removed from group	Removing a user from the Entra group removes them from the JFrog group, but not from the	Pass	Confirmed. A user retaining another group stays on the platform; a user with no remaining groups is disabled and associated tokens are revoked.

ID	Scenario	Expected Result	Pass	Result / Notes
		platform if other memberships remain.		
3.2	App unassignment	Removing a user from the enterprise application disables the user in JFrog.	Pass	Confirmed. For a user with no remaining group memberships, removal from the application changes status to disabled and revokes all associated tokens.
3.3	Manual delete recovery	A user deleted manually in JFrog is re-created on the next sync cycle.	Pass	Confirmed. The next sync re-created the user.
3.4	Cross-group logic	User in Group A and Group B; removed from Group A in Entra remains in Group B and on the platform.	Pass	Confirmed. The user remained active via Group B.

Evidence — removing a SCIM-owned member and the resulting account state:

Edit group 'sum2-group'

Users

174 Available Users

Search

Name	
☐ abhisheks@jfrog.com	⋮
☐ ajayh	⋮
☐ alice-dev	⋮
☐ alpha-slvr-a	⋮
☐ alpha-slvr-b	⋮

4 Selected Users

Search

Name	
☐ anilkt@jfrog.com	⋮
☐ sumanthsv@jfrog.com	⋮
☐ akshathap@jfrog.com	⋮
☐ sumanth	⋮

Group membership after removing one Entra-owned member — remaining members, including local ones, are untouched.

Name ↑	Email	Realm	Groups	Admin	Status	Last Login
☐ jeevanr@jfrog.com	jeevanr@jfrog.com	scim	1 readers	✕	Disabled	

The removed user's account — status Disabled after losing its last in-scope group; not deleted.

De-provisioning model: removing a user from their last in-scope group disables (does not delete) the account and revokes tokens. Disable is reversible by re-adding the user to an in-scope group; no permission targets are lost from the group side.

5 User Identity Matching

Identity matching is by username, not email address. Where multiple accounts share an email but differ in username or realm, SCIM matches only the account whose username equals the value it is sent, and creates a distinct scim-realm account otherwise. The provisioned account holds the group membership and permissions; other same-email accounts hold none.

Name	Email	Realm	Groups	Admin	Status	Last Log
scim-user	sumanthsv@jfrog.com	internal		✗		
sumanth	You sumanthsv@...	internal	4 alpha-ps-admins, sum-gr...	✓		04-07-2020
sumanth2	sumanthsv@jfrog.com	internal		✓		18-03-2020
sumanthsv	sumanthsv@jfrog.com	oauth		✓		23-12-2019
sumanthsv@jfrog.com	sumanthsv@jfrog.com	scim	1 sum2-group	✓		05-05-2020
testtemuser	sumanthsv@jfrog.com	internal		✗		
worksafepermadmin	sumanthsv@jfrog.com	internal		✗		17-03-2020

Multiple accounts sharing one email but distinct usernames and realms; the provisioned account carries the group membership.

This is the primary pre-migration control: align the SCIM username attribute with existing platform usernames, and align the SCIM username with the SAML NameID, before provisioning — otherwise login identities will not inherit provisioned access.

6 Summary of Observed Behaviours

Scenario	Observed behaviour
Same-name group exists	Adopted by name; flipped to External / SCIM-managed; stamped with the Entra object ID
Roles on adopted group	Inherited from the local group (including Platform Administrator)
Membership on adoption	Union — IdP members added, local members preserved
User with same email, different username	Separate provisioned account created; existing accounts ignored
Membership on normal sync	Incremental — only IdP-owned members added or removed
Remove member from group	Member removed from the group on the next sync
Remove member's last in-scope group	Account disabled (not deleted) and tokens revoked;
Manual delete in JFrog	Account re-created on the next sync
Attribute changes in Entra	Overwrite manually-set values in JFrog on the next sync